

Geometric Distribution

The geometric distribution aims to answer the question:

What is the probability that the first success of a series of Bernoulli events occurs in the n^{th} trial?

It might seem complicated, but in fact is very simple. We must only assume that all the $n - 1$ previous events failed, and that the n^{th} one was a success.

$$X \sim \text{geometric}(p) \Rightarrow P[X = n] = (1 - p)^{n-1}p$$

Another possible question is:

What is the probability that it takes *at least* n events before a success occurs?

Now, we are interested in every single geometric distribution case, starting from the success at the n^{th} trial up to infinity.

That said, we can model the problem as such:

$$P[X \geq n] = \sum_{i=n}^{\infty} P[X = i] = \sum_{i=n}^{\infty} (1 - p)^{i-1}p$$

This might seem complicated but, expanding the summation:

$$\sum_{i=n}^{\infty} (1 - p)^{i-1}p = (1 - p)^{n-1}p (1 + (1 - p) + (1 - p)^2 + \dots)$$

Since $(1 - p) < 1$, the right portion of the expression is a geometric sequence, and the infinite summation can be approximated as:

$$S_{\infty} = \frac{a_1}{1 - q} \Rightarrow S_{\infty} = \frac{1}{1 - (1 - p)} = \frac{1}{p}$$

Plugging back in:

$$P[X \geq n] = \frac{(1 - p)^{n-1}p}{p} = (1 - p)^{n-1}$$

Which turns out to be a very simple expression.

References

- [The Geometric Distribution: The First Success of a Bernoulli Distribution](#)